

KaraWell - Thesis Project

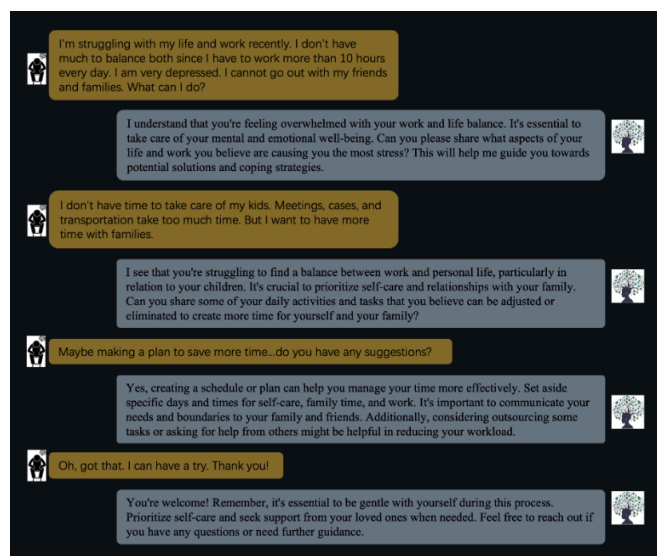
Background: KaraWell is a gamified chatbot that is meant to be a mental health intervention for reducing anxiety levels. In our study, we want to see if the gamification elements we implement in KaraWell directly affect anxiety levels and increase user engagement. To measure its effectiveness as a mental health app, we will use Zung's Self-Rating Anxiety Scale which will measure anxiety levels based on the past few days. For the study, we will have participants answer this scale before and after the study, which will take one week in between. This is meant to see if gamification can directly decrease anxiety levels of our participants.

FUNCTIONAL REQUIREMENTS

The features implemented in the app are defined based on related literature we found on the ideal gamification elements in a mental health intervention.

1. User-bot Interaction

A chat room page is made for users to converse with the chatbot of KaraWell. Using EmoCareAI/ChatPsychiatrist, an AI chatbot known for transforming counseling with personalized support and expert assistance.



The image above is the expected output where ChatPsychiatrist would respond to the user's sentiments in a more personalized way.

2. Gamification elements

a. User streaks

There are two streaks to be recorded in users' activity: the *current streak* is the current number of days where a user has consecutively used the chatroom, which is updated once the user has sent the first

message of the day. The *longest streak* will record the longest consecutive number of days a user can get while using KaraWell's chat room.

b. User badges

The badges are based on the user streaks (ex. One-Day Streak, Three-Day Streak, etc) and are locked when a new user registers into KaraWell. Once streaks reach to 1, 3, 5, 7, 14, and 21, badges will be updated and unlocked for the user to see their progress.

c. Journaling

The journaling feature is meant to be a diary-like feature where users can record what's on their mind for the day. The interface is much similar to the chatroom, but in this case, their notes will be saved in our database for them to view it again. It is then sorted by date and users will see what and how many journals they were able to make per day.

d. Push notifications

To enforce long-term use, and for the upcoming testing phase of the study, push notifications are implemented through a package in Flutter that schedules notifications every 8pm of the day. This is to remind them to use KaraWell and encourage them to continue using it. Notifications will also pop up when the user streak is updated and when a new badge is unlocked.

3. Authentication

For security reasons, we implemented basic authentication for login and signup. As this is just for research purposes, we only require their email and password for authentication.